

Task 1: Key Cybersecurity Terms

Term	One-line definition
Threat	A threat is anything that can potentially cause harm to a computer system, network, or data.
Vulnerability	A vulnerability is a weakness in a system that an attacker could exploit.
Risk	Risk is the possibility of damage or loss when a threat takes advantage of a vulnerability.
Asset	An asset is anything valuable that an organization needs to protect, such as data, servers, applications, or devices.
Attack Vector	An attack vector is the method or path an attacker uses to gain unauthorized access to a system.

Task 2: Real-world Cybersecurity Incident

Incident: IRCTC security and fraudulent-account activity

A recent example involving an Indian digital platform is the security problem surrounding IRCTC's online ticketing system. In 2025, Indian Railways took extensive action against suspicious IRCTC accounts and automated activity. According to reports, approximately 3.03 crore suspicious IRCTC user IDs were deactivated during 2025, while security controls were strengthened to detect bots, brute-force activity and suspicious traffic. Nearly 13,000 suspicious email domains were also blocked.

What happened?

The IRCTC ticketing platform was being targeted by automated scripts/bots and fraudulent accounts, particularly around online and Tatkal ticket bookings. These activities could give malicious or unauthorized users an unfair advantage and potentially affect legitimate customers.

Indian Railways responded by using measures such as:

- Multi-level cybersecurity controls
- CAPTCHA mechanisms
- Aadhaar-based authentication
- Security audits
- Continuous monitoring of Internet traffic
- Blocking suspicious email domains
- Detection of automated/bot activity

Identify the cybersecurity concepts

Threat:

Automated bots, brute-force attempts, and cybercriminals attempting to abuse the ticket-booking system.

Vulnerability:

Weaknesses that can be exploited through automated account creation, scripted requests, or inadequate protection against automated traffic.

Risk:

Unauthorized users could obtain tickets unfairly, legitimate customers could lose access to tickets, and the availability and reliability of the online booking service could be affected.

Conclusion

The IRCTC case demonstrates why large online platforms need continuous monitoring, strong authentication, bot protection, security audits, and mechanisms for detecting suspicious activity.

Source: [Indian Express – Indian Railways cracks down on fake IRCTC IDs](#)

Task 3: Cybersecurity Dashboard

Tool name: Kaspersky Cybermap



Information/metrics you can list from a threat map

Depending on the dashboard you use, you may see information such as:

1. Cyberattack activity – visualization of attacks occurring around the world.
2. Attack origin and destination – countries or regions involved in detected activity.
3. Threat/attack categories – different types of detected cyber threats.

Task 4: Simulated Phishing Email

Example phishing email

Subject: Your Paytm Account Requires Verification

From: Paytm Security <security-alerts@paytm-update24.com>

Reply-To: verify-account@paytm-verify.net

Dear Paytm User,

We have detected suspicious activity on your Paytm account associated with this email address.

Due to multiple failed login attempts and an unusual transaction attempted from an unknown device, your account has been TEMPORARILY SUSPENDED.

To restore access and avoid permanent closure of your account, you must verify your identity within the next 24 HOURS.

👉 Click here to verify your account:
<http://paytm-verification-portal.xyz/secure/verify>

Failure to complete verification will result in your account being permanently disabled and your wallet balance may be forfeited as per our updated policy.

Paytm Customer Support
[This is an automated message. Please do not reply to this email.]

Suspicious elements

Suspicious element	Why it is suspicious
Urgent deadline	Attackers often create urgency so users don't think carefully.
"Unusual activity"	Creates fear and encourages the user to act immediately.
Account suspension threat	Makes the victim worried about losing access.
Unexpected verification request	Legitimate services generally don't ask users to verify accounts through suspicious email links.
Unknown sender domain	example-mail.com is not an official Paytm domain.
Generic greeting	"Dear Customer" instead of the user's name

	can be a warning sign.
Verification button	A phishing button may redirect the victim to a fake login page.

Important: The example uses a fictional/safe sender and does not contain a real credential-harvesting link.

Task 5: AI-Generated Simulated Phishing Email

Fictional Food Delivery App: "FoodiePal"

Subject: Your FoodiePal account will be suspended

From: FoodiePal Account Security <security@foodiepal-example.com>

Hello FoodiePal User,

We detected unusual login activity on your account from a new device.

For your protection, please confirm your account information within 24 hours, otherwise your FoodiePal account may be temporarily suspended.

[Confirm My Account]

If you did not perform this login, please secure your account immediately.

Thank you,

FoodiePal Security Team

Why could this trick users?

1. Fear: The message says the account may be suspended.
2. Urgency: The 24-hour deadline pressures the user to act quickly.
3. Security warning: Mentioning "unusual login activity" makes the email appear legitimate.
4. Call to action: The "Confirm My Account" button encourages the user to click without checking the destination.

5. Impersonation: The sender pretends to be the security team of a familiar service.
6. Generic wording: It does not provide specific information about the user's actual account.

How a user can identify it

Before clicking, the user should:

- Check the sender's email address.
- Hover over links to inspect their destination.
- Avoid entering passwords through links received unexpectedly by email.
- Open the official app or website directly instead.
- Check for spelling, grammar, and unusual wording.
- Report suspicious messages as phishing.